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Agricultural Income Prediction Challenge

Given Dataset & Problem

57K+ Total Records

47,970 training rows and 9,986 unseen test rows, spanning 105 raw columns of uncleaned agricultural data.

Multidimensional Signals

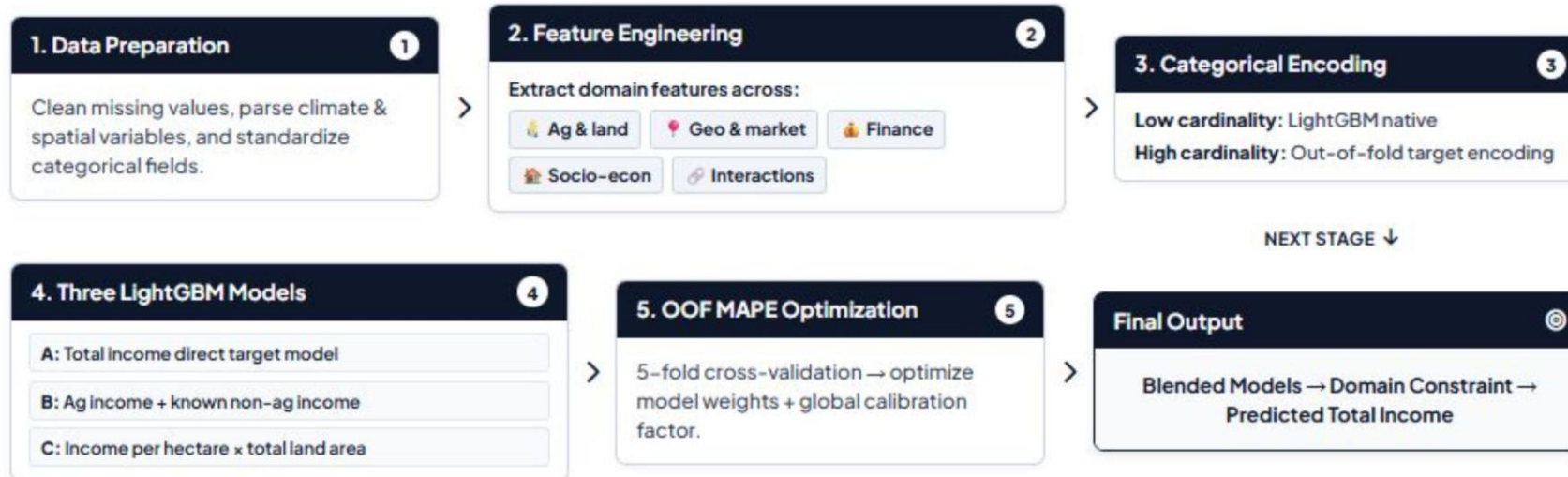
The 105 columns encompass farmer demographics, land and soil characteristics, historical climate data, and socio-economic living indices.

Target & Evaluation

The primary task is predicting Total Income, evaluated on the unseen test set using Mean Absolute Percentage Error (MAPE).

Approach overview

Instead of asking one model to learn income from the raw variables alone, we provide economically meaningful representations of land, geography, climate, market access and financial exposure.



External Data Strategy: Inclusions & Deliberate Exclusions

Macro-Economic Grounding (RBI Data)

The Source: Reserve Bank of India (RBI), Handbook of Statistics on the Indian Economy (Table 10).

The Metric: Per Capita Net State Domestic Product (NSDP) at constant prices.

The Application: Mapped into a baseline **State Economic Tier**, anchoring the model with a reliable proxy for regional wealth.

Geospatial Accessibility (Metro Proximity)

The Source: Hardcoded coordinates for 8 major hubs: Delhi, Mumbai, Kolkata, Chennai, Bengaluru, Hyderabad, Ahmedabad, and Pune.

The Mechanism: Haversine distance to the closest hub, creating **dist_to_nearest_metro_km**.

The Application: Quantifies physical remoteness and market friction, independent of state borders.

Deliberate Exclusion (Minimum Support Price)

The Metric: Minimum Support Price (MSP) data was evaluated but ultimately rejected.

The Roadblock: The core dataset lacks a **Crop Type** column.

The Rationale: Without crop identity, district-level MSP cannot be meaningfully attached or modeled. This remains the highest-priority next step if crop data becomes available.

Novelty in Approach

Domain-Driven Feature Engineering

Our novel improvements to the raw columns given

1

Geographic Isolation (Haversine Formula)

What: Calculates the exact great-circle distance using the Haversine formula from a farmer's coordinates to state capitals and major national metros.

Why: Natively quantifies market access and transport friction, which heavily dictates agricultural profitability.

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Rainfall Variability

What: Rainfall CV = Standard Deviation of Rainfall ÷ Mean Rainfall.

Why:

- Low CV → more consistent rainfall
- High CV → more variable rainfall
- More informative than using only average rainfall

2

Wealth Interaction (Land × Tier Formula)

What: Calculates the product of $\log(1 + \text{land hectares})$ and the RBI-sourced `state_income_tier`.

Why: A non-linear multiplier that scales a farmer's physical asset (land) by the macro-economic baseline (state tier), capturing compound wealth effects.

5

Market Accessibility

What: How easily a farmer can connect their agricultural production to nearby markets.

Market Access = $\text{Road Density} / (1 + \text{Mandi Distance})$

Why: Higher road connectivity → easier movement of produce

- Greater mandi distance → lower accessibility

3

Financial Leverage (Debt per Hectare)

What: Calculates the ratio of `Avg_Disbursement_Amount_Bureau` divided by `Total_Land_For_Agriculture`.

Why: Normalizes absolute loan sizes against productive capacity, acting as a crucial proxy for credit intensity and financial leverage.

6

Land Productivity

What: $\text{Agricultural Land} / \text{Total Land}$

Why:

- Normalizes agricultural area by total landholding
- Distinguishes farmers with similar land sizes but different land utilization

📌 Note: These top 6 signals represent only a fraction of the engineered feature space. The complete end-to-end pipeline engineers dozens of additional features covering climate risk, groundwater balance, census-style living indicators, and crop-season trajectories.

Target Encoding: Managing High-Cardinality Geographies

Mechanism

High-cardinality categorical → Historical target signal

1. Split training data into **5 folds**
2. For each validation fold, calculate the **mean log-income of each category using only the other 4 folds**
3. Apply this encoding to the held-out fold
4. Add a **count feature** showing how much data supports the estimate and how reliable the encoded value is.

The Advantage

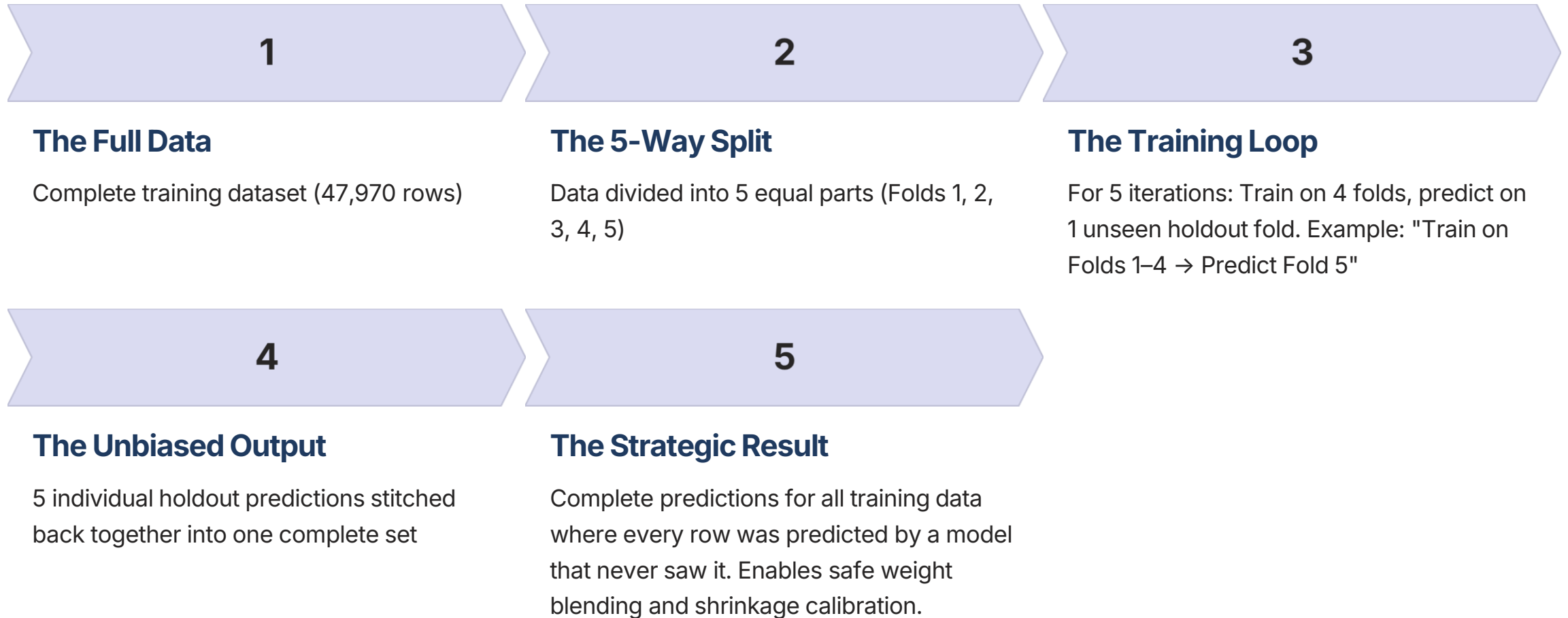
- **Avoids target leakage** — a farmer's own income is never used to encode their validation row
- Handles **5,650+ villages** without thousands of one-hot columns
- Captures **location-specific income patterns**
- **Smoothing** makes rare villages/categories less noisy

The Safeguard

- **5-fold OOF encoding** — each row is encoded using only the other 4 folds.
- **Own target excluded** — a farmer's income never contributes to their own encoded value.
- **Smoothing applied** — rare categories are pulled toward the global mean.
- **Unseen categories → global mean** — prevents unreliable estimates for new locations.

→ see Glossary, Slide 16

Modeling: 5-Fold Out-of-Fold (OOF) Calculation



Modeling Strategy

Three Unique Formulations (The Angles)

- Model A: predicts total income directly.
- Model B: predicts agricultural income, then adds back known non-agri.
- Model C: predicts income per hectare, then multiplies by land size.

The Blend Advantage (Error Cancellation)

Model B is strongest individually:
19.16% MAPE.

But blending all three lowers error to
18.74%.

Each model has a different
bias/variance trade-off, so their blind
spots help cancel out.

Calibration & Logic Floor (The Final Polish)

MAPE shrinkage: multiply the blended
output by **0.9675** to better match the
scoring objective.

Hard floor: predictions are floored at
known non-agricultural income.

This keeps total income realistic: it
cannot fall below its base non-agri
component.

Model Selection: Why LightGBM?

Native Categorical Power

The Advantage: Avoids the "dimensionality curse" of one-hot encoding.

- **Low-cardinality features** such as **State, Gender, Marital Status, Ownership & Soil Type** have a limited number of categories.
- LightGBM can **handle these categories directly**, avoiding unnecessary one-hot columns.
- The model can learn **different splits and patterns for categories** rather than treating them as continuous numerical values.
- This keeps the feature space **compact and efficient** while preserving categorical information.

Non-Linear Relationships

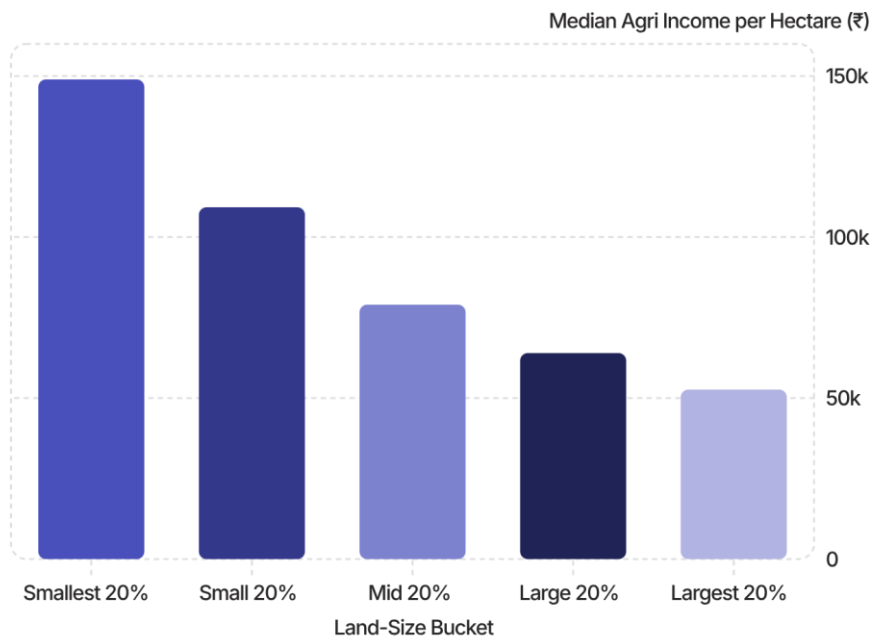
- **Tree-based learning** → LightGBM does not assume a straight-line relationship between features and income.
- **Threshold effects** → Learns patterns such as *income changing differently after a certain land size or market distance*.

Controlling Overfitting

Feature sampling (75%) → Each tree sees only a subset of features.

- **Minimum leaf size (35)** → Prevents trees from creating tiny, highly specific groups.
- **L2 regularization ($\lambda = 3$)** → Penalizes overly complex tree solutions.

Sanity Check: The Inverse Farm-Size Relationship



The Organic Discovery

The model organically learned that per-hectare income drops as farm size increases. We did not hard-code this rule; it emerged naturally from the data.

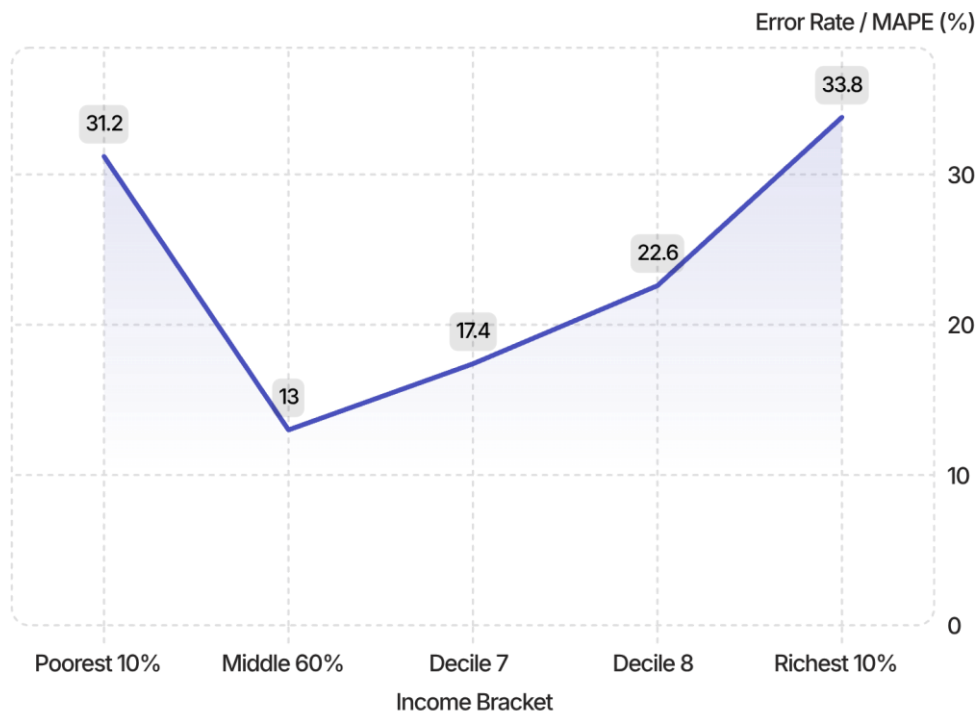
The Real-World Anchor

This perfectly matches the "inverse farm size-productivity relationship" in agricultural economics. Smallholders farm their plots more intensively with higher labor input, while larger farms scale with lower per-hectare intensity.

Proof of Genuine Learning

Reproducing known economic laws is stronger proof of a robust model than a standalone MAPE score. We also confirmed 0 violations of the non-agri income floor across all 9,986 test rows.

Sanity Check: The U-Shaped Error Distribution



Middle 60% is the sweet spot: dense training data and moderate variance keep errors low.

The Poorest Decile (31.2% Error)

This high percentage error is largely a mathematical artifact. The MAPE metric heavily penalizes small absolute rupee errors when the baseline income is very low.

The Middle Class Sweet Spot (12-14% Error)

With dense training data and moderate variance, the model confidently captures the core economic patterns for the vast majority of farmers (incomes between ₹680K – ₹1.15M).

The Richest Decile (33.8% Error)

This bracket suffers from extreme variance and data sparsity. Training examples above ₹6.3M are thin, containing heterogeneous outliers reporting incomes up to ₹8 crore.

Honest Limitations & Next Steps

1

Hyperparameter Optimization

- **Current State:** Relies on manual tuning for learning rate, leaves, and regularization.
- **Next Step:** Implement a systematic Bayesian/Optuna sweep to extract fractional performance gains automatically.

1

Validation Optimism

- **Current State:** Blend weights and the shrinkage factor were fit and evaluated on the same 5-fold Out-of-Fold dataset.
- **Next Step:** Implement nested cross-validation. This will provide a stricter, less optimistic estimate than our current 18.74% best-case score.

1

The 'Crop Type' Roadblock

- **Current State:** External Minimum Support Price (MSP) and crop yield data were abandoned.
- **Next Step:** Request a "Crop Type" column from the organizers. Without knowing what a farmer is growing, macro-agricultural data cannot be joined or modeled effectively.

Key Insights

Interactions Over Raw Data

- Observation: The land × income tier feature outperforms both variables individually.
- Takeaway: Carefully engineered interactions matter significantly more than raw dataset features.

Model Blending Superiority

- Observation: While our best single LightGBM formulation achieved a 19.16% MAPE, blending three unique mathematical formulations dropped the overall error to 18.74%
- Takeaway: Averaging multiple models forces their unique blind spots to cancel each other out, yielding significantly better accuracy and robustness than relying on a single model.

Evidence of Genuine Learning

- Observation: Test predictions organically reproduce the inverse farm-size/productivity relationship without being prompted.
- Takeaway: This confirms the model captured real-world economic signals, rather than overfitting or memorizing dataset noise.

High-Cardinality Transformation

- Observation: Out-of-fold target encoding for the DISTRICT column yielded the highest feature gain.
- Takeaway: Safely translating messy geographic strings into numeric economic baselines is the single most powerful feature engineering step in the pipeline.

Real-World Applications

Loan Sizing

The Application: Estimate repayment capacity for farmers who have no formal income records.

KCC Eligibility

The Application: Screen Kisan Credit Card (KCC) applications much faster in scenarios where traditional documentation is thin.

Rural Market Sizing & FMCG Expansion

The Application: Enables strategy teams and FMCG brands to map granular rural purchasing power for targeted product expansion.

Smart Farming Platforms Integration

The Application: Combine this income prediction engine with existing smart farming tools—such as soil health analysis, crop disease detection, and mandi price forecasting—to offer a holistic, data-driven financial planning platform for farmers.

Glossary of Terms

ECONOMIC TERMS:

- **Financial leverage** — debt level relative to the assets (land) a farmer holds
- **Inverse farm size–productivity relationship** — a documented pattern where smaller farms often produce more income per hectare than larger ones
- **Land Holding Index (LHI)** — average agricultural land per person in a given village
- **Market access** — how easily a farmer can reach buyers/markets to sell produce
- **NSDP (Net State Domestic Product)** — a state's average economic output per person, used as our wealth-tier proxy
- **Remoteness proxy** — a distance-based stand-in for how well-connected a farmer is to markets/cities

MODELING/STATISTICS TERMS:

- **Ensemble/blending** — combining multiple models' predictions to cancel out individual errors
- **Feature engineering** — turning raw/messy data into clean, meaningful inputs for a model
- **Haversine formula** — the standard formula for real-world distance between two GPS coordinates
- **MAPE** — Mean Absolute Percentage Error, average prediction error as a %, our scoring metric
- **Out-of-fold (OOF)** — calculating a value without letting the model see the answer for that row, preventing cheating
- **Overfitting** — when a model memorizes noise in training data instead of general patterns
- **Shrinkage/calibration** — a small correction applied to predictions to better match the scoring metric
- **Target encoding** — replacing a category (e.g. village name) with a number representing its typical outcome